

1. Command: **cat**

1.1 **cat file.txt**

Description: Displays the contents of a file.

```
cat hello.txt
```

Output:

```
Hello OS Lab
```

1.2 **cat -n file.txt**

Description: Displays file contents with line numbers.

```
cat -n hello.txt
```

Output:

```
1 Hello OS Lab
```

1.3 **cat > file.txt**

Description: Creates and writes text into a file.

```
cat > notes.txt
```

1.4 **cat file1 file2**

Description: Combines and displays multiple files.

```
cat file1.txt file2.txt
```

1.5 `cat -E file.txt`

Description: Shows `$` at end of each line.

```
cat -E hello.txt
```

2. Command: `ls`

2.1 `ls`

Description: Lists files and directories.

```
ls
```

2.2 `ls -l`

Description: Displays files in long listing format.

```
ls -l
```

2.3 `ls -a`

Description: Shows hidden files.

```
ls -a
```

2.4 `ls -lh`

Description: Shows file sizes in human-readable format.

```
ls -lh
```

2.5 `ls -R`

Description: Lists directories recursively.

ls -R

3. Command: **cp**

3.1 **cp source destination**

Description: Copies a file.

cp hello.txt copied.txt

3.2 **cp -r dir1 dir2**

Description: Copies directories recursively.

cp -r folder1 folder2

3.3 **cp -i**

Description: Prompts before overwrite.

cp -i file1.txt file2.txt

3.4 **cp -v**

Description: Shows copying process.

cp -v hello.txt copied.txt

3.5 **cp *.txt backup/**

Description: Copies all **.txt** files.

```
cp *.txt backup/
```

4. Command: touch

4.1 touch file.txt

Description: Creates empty file.

```
touch hello.txt
```

4.2 touch file1 file2

Description: Creates multiple files.

```
touch a.txt b.txt
```

4.3 touch -a file.txt

Description: Updates access time only.

```
touch -a hello.txt
```

4.4 touch -m file.txt

Description: Updates modification time only.

```
touch -m hello.txt
```

4.5 touch -t

Description: Sets custom timestamp.

```
touch -t 202605201200 hello.txt
```

5. Command: **rm**

5.1 **rm file.txt**

Description: Deletes a file.

```
rm copied.txt
```

5.2 **rm -i file.txt**

Description: Asks before deletion.

```
rm -i hello.txt
```

5.3 **rm -r folder**

Description: Deletes directory recursively.

```
rm -r testfolder
```

5.4 **rm -f file.txt**

Description: Force deletes file.

```
rm -f hello.txt
```

5.5 **rm -rf folder**

Description: Force deletes folder recursively.

```
rm -rf os-lab
```

6. Command: **mkdir**

6.1 **mkdir** folder

Description: Creates a directory.

```
mkdir newlab
```

6.2 **mkdir** dir1 dir2

Description: Creates multiple directories.

```
mkdir lab1 lab2
```

6.3 **mkdir** -p

Description: Creates nested directories.

```
mkdir -p os/lab/week1
```

6.4 **mkdir** -v

Description: Shows directory creation process.

```
mkdir -v project
```

6.5 **mkdir** foldername

Description: Creates folder in current directory.

```
mkdir practice
```

7. Command: **pwd**

7.1 **pwd**

Description: Shows current directory.

```
pwd
```

7.2 **pwd -P**

Description: Shows physical directory path.

```
pwd -P
```

7.3 **pwd -L**

Description: Shows logical directory path.

```
pwd -L
```

7.4 **echo \$(pwd)**

Description: Prints working directory using command substitution.

```
echo $(pwd)
```

7.5 **cd /home && pwd**

Description: Changes directory and prints path.

```
cd /home && pwd
```

8. Command: **grep**

8.1 **grep word file.txt**

Description: Searches text in file.

```
grep main hello.txt
```

8.2 **grep -i**

Description: Case-insensitive search.

```
grep -i hello notes.txt
```

8.3 **grep -n**

Description: Shows line numbers.

```
grep -n int program.c
```

8.4 **grep -r**

Description: Recursive search.

```
grep -r main .
```

8.5 **grep -v**

Description: Displays lines NOT matching pattern.

```
grep -v hello notes.txt
```